

A Phase Symmetry Conjecture for the Riemann Zeta Function

André Gualberto

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Abstract

We examine the condition $K(s)K(1-s) \in \mathbb{R}$ for $K(s) = 2^s - 1$, which is algebraically equivalent to $\Re(s) = 1/2$ or $\Im(s) = n\pi/\ln 2$ for integer n . If a non-trivial zero of the Riemann zeta function were to lie off the critical line, it could only do so on one of these exceptional lines. We present a numerical scan of 200 such lines that finds no candidate. The exclusion of these lines remains an open problem.

1 Introduction

The fundamental equation

$$\boxed{\frac{1}{2} \times 2 = 1}$$

is the algebraic seed of the symmetry explored here. The number 2 is the base of the parity alternation that separates odd primes from even numbers; $1/2$ is the critical line of the Riemann hypothesis; and 1 is the primordial unit.

The valence function $K(s) = 2^s - 1$ captures the role of the base 2 in the functional equation of $\zeta(s)$. The condition $K(s)K(1-s) \in \mathbb{R}$ forces $\Re(s) = 1/2$ unless $\Im(s) = n\pi/\ln 2$.

2 Algebraic Symmetry

Define $K(s) = 2^s - 1$.

Proposition 1. $K(s)K(1-s) \in \mathbb{R}$ iff $\Re(s) = \frac{1}{2}$ or $\Im(s) = n\pi/\ln 2$ for some $n \in \mathbb{Z}$.

Proof. Write $s = \sigma + it$. Then

$$K(s)K(1-s) = (2^s - 1)(2^{1-s} - 1) = 3 - 2^s - 2^{1-s}.$$

Since $2^{1-s} = 2 \cdot 2^{-s}$,

$$\Im(K(s)K(1-s)) = -\Im(2^s + 2^{1-s}).$$

With $2^s = e^{\sigma \ln 2} e^{it \ln 2}$,

$$\begin{aligned} \Im(2^s + 2^{1-s}) &= e^{\sigma \ln 2} \sin(t \ln 2) + e^{(1-\sigma) \ln 2} \sin(-t \ln 2) \\ &= (e^\sigma - e^{1-\sigma}) \ln 2 \sin(t \ln 2). \end{aligned}$$

This vanishes iff $\sigma = 1/2$ or $t \ln 2 = n\pi$. □

3 Conjecture

Based on the algebraic and numerical evidence, we propose:

If $\zeta(s) = 0$ and $0 < \Re(s) < 1$, then $K(s)K(1-s) \in \mathbb{R}$.

This conjecture is equivalent to the Riemann Hypothesis, since the phase condition $K(s)K(1-s) \in \mathbb{R}$ forces $\Re(s) = 1/2$ (except for the exceptional lines $t = n\pi / \ln 2$, which are numerically excluded but await a rigorous analytic proof).

4 Numerical Evidence

We scanned 200 exceptional lines ($n = 1, \dots, 200$, $\sigma = 0.05, \dots, 0.95$, step 0.05) using mpmath with 30-digit precision. No candidate with $|\zeta(s)| < 10^{-4}$ was found. This evidence is consistent with the conjecture but does not constitute a proof.

5 Conclusion

The conjecture is open. A proof of the remaining step—excluding the exceptional lines analytically—would prove the Riemann Hypothesis.